

Technical Documentation

per Annex IV, Regulation (EU) 2024/1689 (EU AI Act)

Estimate the calibrated probability of heart disease from 11 clinical features

Provider: Chinmay Nighojkar (portfolio project)

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Risk classification (summary)

Classification (as built): Minimal-risk (no specific obligations)

Primary citations: Article 6(2); Annex III area 5 (adjacent, not triggered)

Conclusion: As built – an educational demonstrator not placed on the EU market and not connected to any Annex III deployment – the system is not high-risk today. HOWEVER, its intended-purpose class (clinical risk scoring) is exactly what Annex III area 5 and Article 6(1) capture: the moment it were used for benefit eligibility, insurance pricing, triage, or as a medical device, it would become high-risk and the full Chapter III obligations would attach.

Annex IV(1) – General description of the AI system

Intended purpose: Estimate the calibrated probability of heart disease from 11 clinical features. Built as an educational MLOps demonstrator; NOT a medical device and not for clinical use.

Provider / version: Chinmay Nighojkar (portfolio project) / v1.2

Relation to previous versions: Version 1.2. Model selected by 5-fold CV ROC-AUC over RandomForest, GradientBoosting and XGBoost.

Interactions (b): Served via a FastAPI /predict endpoint (api/main.py); a Streamlit dashboard (dashboard/app.py) reads model metadata, the prediction log and drift reports. No external AI systems are involved.

Software versions (c): scikit-learn 1.5.2, xgboost 2.1.3, shap 0.46.0, evidently 0.4.40, fastapi 0.115.6 (see requirements.txt).

Forms placed on market (d): Not placed on any market. Runs as a containerised service (Dockerfile) and a local dashboard for demonstration only.

Target hardware (e): Commodity CPU; no GPU required for inference.

User interface (g): FastAPI JSON API and a Streamlit monitoring dashboard.

Instructions for use (h): Submit 11 clinical features to /predict; the response returns a calibrated risk_score, a 0/1 decision at a 0.5 threshold, and per-feature SHAP attributions. Output is decision-support only and must be reviewed by a human.

Annex IV(2) – Detailed description of the elements of the AI system and of its development process

Development methods (a): Supervised learning on the combined UCI heart-disease dataset. GridSearchCV hyper-parameter tuning on ROC-AUC; probability calibration with CalibratedClassifierCV (sigmoid/Platt, 5-fold).

Design specifications (b): RandomForestClassifier (n_estimators=400, max_depth=None, min_samples_leaf=10). The uncalibrated tree is retained only to compute SHAP attributions; the calibrated classifier produces the served probability.

Optimisation target (b): ROC-AUC during selection/tuning; calibrated probability quality (Brier score) for the served output, because a risk score must be trustworthy as a probability, not just rank-ordered.

System architecture (c): preprocessing.py -> train.py (model + calibration + SHAP model persisted to models/) -> FastAPI serving with prediction logging -> monitor.py (Evidently drift) -> Streamlit dashboard.

Computational resources (c): Trained on CPU in seconds; artefacts persisted with joblib.

Data requirements (d): Combined UCI heart-disease dataset: 918 rows after de-duplication, split 734 train / 184 test (stratified, random_state=42). Features: age, sex, chest_pain_type, resting_bp_s, cholesterol, fasting_blood_sugar, resting_ecg, max_heart_rate, exercise_angina, oldpeak, st_slope.

Data cleaning (d): Values of 0 for cholesterol (18.7% of rows) and resting_bp_s encode a MISSING measurement, not a real reading, and leak a spurious signal (rows with missing cholesterol show an 88% disease rate vs 55% overall). These zeros are replaced with the training-set median (cholesterol=238.0, resting_bp_s=130.0).

Human-oversight assessment (e): Per-prediction SHAP attributions let a human reviewer see which features drove each score; the model card documents that output is decision-support only. No autonomous decision is taken.

Pre-determined changes (f): Retraining is performed by re-running train.py on updated data; there is no automated drift-triggered retraining (documented limitation).

Validation & testing (g): 5-fold stratified cross-validation for selection/tuning; a held-out 184-row test set for final metrics; automated tests in tests/ for the API and monitor.

Cybersecurity (h): Input validation via Pydantic request schemas on the API; pinned dependencies; containerised runtime. No authentication layer (demonstrator).

Accuracy metrics (g):

Metric	Value
roc_auc	0.9273
accuracy	0.8913
precision	0.8868
recall	0.9216
f1	0.9038
brier_calibrated	0.0980
brier_uncalibrated	0.1023

Subgroup metrics by sex (g – discriminatory-impact check):

Group	n	Accuracy	ROC-AUC	Recall
female	38	0.9474	0.9479	0.8333
male	146	0.8767	0.9056	0.9271

Annex IV(3) – Monitoring, functioning and control of the AI system

Capabilities & limitations: Capability: calibrated heart-disease risk from tabular clinical inputs. Limitation: trained on a small public dataset, not validated on any real population; performance may not transfer.

Per-group accuracy: By sex – female (n=38): ROC-AUC 0.9479, recall 0.8333; male (n=146): ROC-AUC 0.9056, recall 0.9271. The smaller female subgroup warrants caution.

Overall expected accuracy: Held-out ROC-AUC 0.9273, accuracy 0.8913, recall 0.9216 at a 0.5 threshold.

Foreseeable unintended outcomes: False negatives (missed disease) are the most consequential error for a screening-style score; recall is reported alongside accuracy for this reason. Distribution shift in the patient population would degrade performance.

Human-oversight measures: SHAP explanations per prediction; documented intended-use boundary; the dashboard exposes model health and drift to an overseer.

Input-data specifications: 11 numeric/categorical clinical features with the ranges and encodings documented in the model card; the API rejects malformed inputs.

Annex IV(4) – Appropriateness of the performance metrics

ROC-AUC captures ranking quality independent of threshold and is standard for clinical risk models. Brier score and a reliability curve are reported because a probability that will be read by a human must be calibrated, not merely discriminative. Recall is highlighted because missed disease is the costliest error. Subgroup metrics test for disparate accuracy (Art. 10/15).

Annex IV(5) – Risk management system (Article 9)

Iterative, documented risk management: risks and mitigations are recorded in docs/decision_log.md, and residual risks are stated openly rather than closed out. Mapped article-by-article in the requirement->evidence map.

Identified risks:

- Data leakage from missing-value sentinels (cholesterol/resting_bp_s == 0).
- Disparate accuracy across sex subgroups (smaller female sample).
- Silent performance decay under population/concept drift.
- Over-reliance on an uncalibrated score by a human reader.

Mitigations:

- Median imputation of sentinel zeros; documented in preprocessing.py and the model card.
- Subgroup metrics measured and reported by sex.
- Evidently input-drift monitoring with a drift-simulation demonstration.
- Probability calibration (Platt) + per-prediction SHAP for interpretability.

Residual risks (stated openly):

- Article 9: Risk identification and mitigation are documented and reproducible, but retraining is manual (src/train.py); there is no automated, drift-triggered risk-response loop.
- Article 14: Oversight is supported by explanation and documentation, but there is no formal override workflow or logged human-in-the-loop sign-off, since the system is a portfolio demonstrator, not a deployed clinical tool.
- Article 15: Monitoring detects INPUT (data) drift only. Performance/concept drift is not detected because ground-truth outcomes for past predictions are not collected – a documented Article 15 / Article 72 limitation, not a solved control.
- Article 72: Covers input-distribution monitoring; a full Art. 72 plan would add outcome collection, performance tracking and a documented response procedure.

Annex IV(6) – Lifecycle changes

Current model version 1.2. Version history and the rationale for each change are maintained in docs/decision_log.md. This compliance layer adds a tamper-evident audit trail over the prediction log.

Annex IV(7) – Harmonised standards applied

Harmonised standards applied: None formally applied (portfolio demonstrator).

Solutions where no standard: In the absence of applied harmonised standards, conformity-relevant practices follow widely used references in spirit: model cards (Mitchell et al., 2019) for transparency, calibration/Brier for accuracy, and SHAP for explainability. ISO/IEC 42001 and the forthcoming CEN-CENELEC AI Act harmonised standards would be the reference points for a real deployment.

Annex IV(8) – EU declaration of conformity

Status: Not applicable – demonstrator, not placed on the EU market.

No EU declaration of conformity (Art. 47) is issued because this is a portfolio artifact and not a high-risk AI system placed on the market. The field is retained to show where a real Annex IV(8) copy would be attached.

Annex IV(9) – Post-market monitoring (Article 72)

System description: Evidently-based input-drift monitoring over the live prediction log (src/monitor.py), producing dated HTML reports in monitoring/reports/.

Monitoring plan: Run drift reporting on a schedule and on demand; investigate flagged feature drift; a –simulate mode demonstrates the report firing on a shifted population.

Known limitations: Input drift only. A complete Art. 72 plan would add outcome collection, performance-drift tracking, alert thresholds and a documented response SLA.